

A sample $X_1, \dots, X_n$ is observed from $Geometric(\theta)$ distribution. Derive the Bayes estimator of θ under the $Beta(\alpha, \beta)$ prior distribution and the squared-error loss function.

SOLUTION.

$$\begin{aligned} f(\mathbf{X}|\theta) &= f(X_1, \dots, X_n|\theta) = \prod_{i=1}^n (1-\theta)^{X_i-1} \theta = (1-\theta)^{\sum X_i - n} \theta^n, \\ \pi(\theta) &\sim \theta^{\alpha-1} (1-\theta)^{\beta-1}, \quad 0 < \theta < 1. \end{aligned}$$

Find the posterior density of θ ,

$$\pi(\theta | \mathbf{X}) \sim f(\mathbf{X}|\theta)\pi(\theta) \sim (1-\theta)^{\sum X_i - n} \theta^n \cdot \theta^{\alpha-1} (1-\theta)^{\beta-1} = \theta^{\alpha+n-1} (1-\theta)^{\beta+\sum X_i - n-1},$$

which is $Beta(\alpha_x, \beta_x)$, with

$$\begin{aligned} \alpha_x &= \alpha + n, \\ \beta_x &= \beta + \sum X_i - n. \end{aligned}$$

(so, Beta family is conjugate to Geometric).

Then,

$$\mathbf{E}(\theta | \mathbf{X}) = \frac{\alpha_x}{\alpha_x + \beta_x} = \frac{\alpha + n}{\alpha + \beta + \sum X_i}.$$

Inventory of some distributions

Discrete distributions					
Distribution	Parameters	Probability mass function	$\mathbf{E}(X)$	$\text{Var}(X)$	Comments
Binomial $\mathcal{B}(n, p)$	$n = 1, 2, \dots$ $p \in (0, 1)$	$\binom{n}{x} p^x (1-p)^{n-x},$ $x = 0, 1, \dots, n$	np	$np(1-p)$	
Geometric $\mathcal{G}e(p)$	$p \in (0, 1)$	$(1-p)^{x-1} p, \ x = 1, 2, \dots$	$1/p$	$(1-p)/p^2$	$\mathcal{NB}(1, p)$
Negative Binomial $\mathcal{NB}(k, p)$	$k = 1, 2, \dots$ $p \in (0, 1)$	$\binom{x-1}{r-1} p^k (1-p)^{x-k},$ $x = k, k+1, \dots$	$1/p$	$(1-p)/p^2$	
Poisson $\mathcal{P}(\lambda)$	$\lambda > 0$	$e^{-\lambda} \lambda^x / x!, \ x = 0, 1, 2, \dots$	λ	λ	
Continuous distributions					
Distribution	Parameters	Density	$\mathbf{E}(X)$	$\text{Var}(X)$	Comments
Normal $\mathcal{N}(\mu, \sigma)$	$\mu \in R$ $\sigma > 0$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \ x \in R$	μ	σ^2	
Uniform $\mathcal{U}(a, b)$	$a < b$	$\frac{1}{b-a}, \ a < x < b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$Be(1, 1)$
Gamma $\mathcal{G}(\alpha, \beta)$	$\alpha, \beta > 0$	$\frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$ $x > 0$	$\alpha\beta$	$\alpha\beta^2$	
Exponential $\mathcal{E}(\beta)$	$\beta > 0$	$\frac{1}{\beta} e^{-x/\beta}, \ x > 0$	β	β^2	$\mathcal{G}(1, \beta)$
Chi-square $\chi^2(\nu)$	$\nu > 0$ (d.f.)	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}$ $x > 0$	ν	2ν	$\mathcal{G}(\nu/2, 2)$
Beta $\mathcal{B}e(\alpha, \beta)$	$\alpha, \beta > 0$	$\frac{\Gamma(\alpha, \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$ $0 < x < 1$	$\frac{\alpha}{\alpha + \beta}$	$\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$	
Cauchy $\mathcal{C}(\alpha, \beta)$	$\alpha \in R$ $\beta > 0$	$\frac{\beta}{\pi[\beta^2 + (x - \alpha)^2]}, x \in R$	—	—	
Inverse Gamma $\mathcal{IG}(\alpha, \beta)$	$\alpha, \beta > 0$	$\frac{1}{\Gamma(\alpha)\beta^\alpha x^{\alpha+1}} e^{-1/x\beta}$ $x > 0$	$\frac{1}{\beta(\alpha - 1)}$ if $\alpha > 1$	$\frac{1}{\beta^2(\alpha - 1)^2(\alpha - 2)}$ if $\alpha > 2$	
Pareto $\mathcal{P}a(x_0, \alpha)$	$\alpha, x_0 > 0$	$\frac{\alpha x_0^\alpha}{x^{\alpha+1}}, \ x > x_0$	$\frac{\alpha x_0}{\alpha - 1}$ if $\alpha > 1$	$\frac{\alpha x_0^2}{(\alpha - 1)^2(\alpha - 2)}$ if $\alpha > 2$	